COURSE 1 · MATHEMATICS FOUNDATIONS

Mathematics foundations

Theory reader, audio edition

Number sense, units, geometry, trigonometry, statistics, vectors
and the habits of mathematical thinking: the toolkit every other course uses.

7 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 1.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

1.1 Numbers and arithmetic fluency

1.2 Units, measurement and dimensional analysis

1.3 Geometry

1.4 Trigonometry

1.5 Probability and statistics

1.6 Vectors

1.7 Mathematical thinking

This course rebuilds the quantitative toolkit that sits under every other course in the package. Much of it will feel familiar once dusted off. The aim is fluency: being able to estimate, convert, and reason with numbers quickly and confidently, so that the physics and chemistry later can be about ideas rather than arithmetic.

Module 1.1

Numbers and arithmetic fluency

Science deals in numbers that are absurdly large or vanishingly small. The tools for handling them are fractions, percentages, powers of ten and, above all, the habit of estimating before you calculate.

By the end of this module you can

- Work comfortably with fractions, ratios and percentages
- Read, write and calculate in scientific notation
- Estimate answers to within a factor of ten and judge significant figures

Fractions, ratios and percentages

A fraction, a ratio and a percentage are three ways of saying the same thing. Three quarters, three to one with four parts in total, and seventy-five per cent all describe the same share. Moving between them is the first fluency to rebuild.

The one percentage idea that trips adults up is the difference between percentage points and per cent. If unemployment rises from four per cent to five per cent, it has risen by one percentage point, but by twenty-five per cent of its starting value. You will have met this in reading research. Science uses the same distinction constantly.

Percentage change is always measured against the starting value: the change divided by the original, times one hundred [Equation 1.1-1]. A price that falls by fifty per cent and then rises by fifty per cent ends up at seventy-five per cent of where it began, because the second change is measured from a smaller base.

Powers of ten and scientific notation

Writing the mass of the Earth as six followed by twenty-four zeros is unreadable. Scientific notation writes it as six times ten to the power twenty-four. The first part, called the mantissa or coefficient, is a number from one up to, but not including, ten. The power of ten carries the size.

Multiplying numbers in scientific notation means multiplying the coefficients and adding the exponents [Equation 1.1-2]. Dividing means dividing the coefficients and subtracting the exponents [Equation 1.1-3]. Three times ten to the eight, multiplied by two times ten to the minus three, is six times ten to the five.

Negative exponents describe small things. Ten to the minus three is one thousandth. A nanometre is ten to the minus nine metres. The metric prefixes, kilo, mega, giga for big and milli, micro, nano for small, are simply names for powers of ten in steps of a thousand, and they are listed in the Visual Companion.

Estimation and significant figures

Physicists prize the order-of-magnitude estimate: getting an answer right to the nearest power of ten using rounded numbers. How many piano tuners are there in Brisbane? Population about two and a half million, perhaps one piano per twenty households, a tuning once a year, a tuner managing about a thousand tunings a year. Roughly fifty. The method matters more than the answer, because it exposes which assumptions drive the result.

Significant figures express how precisely you know something. A length measured as two point five zero metres claims three significant figures; two point five claims two. When you multiply or divide, the answer should carry no more significant figures than the least precise input. Reporting a calculated result to eight decimal places when the data had two is a classic sign of a beginner, and examiners notice.

Bridge from the social sciences

Percentage points versus per cent, relative risk versus absolute risk, and base effects in growth rates are everyday concerns in social research. The same distinctions govern error, uncertainty and change in the sciences.

Key ideas to carry forward

- Percentage change is measured against the starting value
- In scientific notation, multiply coefficients and add exponents
- Estimate first; it tells you whether a calculated answer is sensible
- Report results to the precision of your least precise input

Module 1.2

Units, measurement and dimensional analysis

A number without a unit is meaningless in science. Units are not decoration: they can be multiplied, divided and cancelled like algebra, and they will catch more of your mistakes than any other check.

By the end of this module you can

- Use SI base units and convert between units with conversion factors
- Check an equation by dimensional analysis
- Describe measurement uncertainty and combine it simply

The SI system

Science runs on the International System of Units, known by its French initials SI. Seven base units cover everything: the metre for length, kilogram for mass, second for time, ampere for electric current, kelvin for temperature, mole for amount of substance, and candela for luminous intensity. Every other unit is built from these. A newton, the unit of force, is a kilogram metre per second squared [Equation 1.2-1]. A joule, the unit of energy, is a newton metre.

This matters because equations only work if the units are consistent. Put kilometres into an equation that expects metres and you will be wrong by a factor of a thousand. The safest habit is to convert everything to SI base units before you substitute.

Converting with conversion factors

A conversion factor is a fraction whose top and bottom are equal, such as one thousand metres over one kilometre. Because it equals one, multiplying by it changes the units without changing the quantity. To convert seventy-two kilometres per hour into metres per second, multiply by one thousand metres per kilometre and by one hour per three thousand six hundred seconds. Kilometres cancel, hours cancel, and you are left with twenty metres per second.

Areas and volumes need care. One metre is one hundred centimetres, but one square metre is ten thousand square centimetres, and one cubic metre is a million cubic centimetres [Equation 1.2-2]. The conversion factor is raised to the same power as the unit.

Dimensional analysis

Every physical quantity has dimensions: combinations of mass, length, time and so on. Speed has dimensions of length divided by time. Acceleration is length divided by time squared. A valid equation must have the same dimensions on both sides, and you can only add or subtract quantities with the same dimensions. You cannot add a speed to a mass.

This gives a powerful check. If you derive a formula for a pendulum's period and the right side comes out in metres, something is wrong, because a period is a time. Dimensional analysis can even

suggest the form of a law. If a pendulum's period depends only on its length and on gravity, the only combination with dimensions of time is the square root of length over gravity [Equation 1.2-3]. The full law adds a factor of two pi, which dimensions alone cannot give.

Measurement and uncertainty

No measurement is exact. A ruler marked in millimetres gives lengths to about half a millimetre either way. We write a result as the best value plus or minus its uncertainty, for example twelve point four plus or minus zero point one centimetres. The percentage uncertainty is the uncertainty divided by the value [Equation 1.2-4].

When you multiply or divide measured quantities, a simple rule of thumb is to add their percentage uncertainties. When you add or subtract, add the absolute uncertainties. Systematic errors, like a stretched tape measure, shift every reading the same way and do not shrink with repetition. Random errors scatter readings and do shrink when you average.

Bridge from the social sciences

Random versus systematic error maps directly onto variance versus bias in survey methods. Repeating a biased measurement more times makes you more confident in the wrong answer.

Key ideas to carry forward

- Convert to SI base units before substituting
- Conversion factors equal one; raise them to the power of the unit
- Both sides of an equation must have the same dimensions
- Averaging reduces random error, not systematic error

Module 1.3

Geometry

Geometry is the mathematics of shape and space. A handful of results, Pythagoras, similar triangles, and the formulas for circles and spheres, do almost all the work you will need in physics and astronomy.

By the end of this module you can

- Use Pythagoras' theorem in two and three dimensions
- Use similar triangles and scale factors, including for area and volume
- Calculate perimeters, areas, surface areas and volumes of standard shapes

Angles and triangles

Angles in a triangle add to one hundred and eighty degrees. Angles around a point add to three hundred and sixty. Where a line crosses two parallel lines, matching angles are equal. These facts let you find unknown angles in almost any diagram.

A right-angled triangle obeys Pythagoras' theorem: the square on the longest side, the hypotenuse, equals the sum of the squares on the other two sides [Equation 1.3-1]. A triangle with sides three, four and five is the classic example. In three dimensions the same idea extends: the diagonal of a box is the square root of the sum of the squares of its three sides [Equation 1.3-2].

Similarity and scale

Two shapes are similar if one is an enlargement of the other: same angles, all lengths in the same ratio. That ratio is the scale factor. Similar triangles are how surveyors measured mountains and how Eratosthenes measured the Earth more than two thousand years ago.

Here is the idea that matters most in science. If lengths scale by a factor k, areas scale by k squared and volumes by k cubed [Equation 1.3-3]. Double the size of an animal and its weight, which goes with volume, grows eightfold, while the strength of its bones, which goes with cross-sectional area, only grows fourfold. This square-cube law is why there are no insects the size of horses, and it reappears in heat loss, cell size and star formation.

Circles and spheres

A circle of radius r has circumference two pi r and area pi r squared [Equation 1.3-4]. Pi is the ratio of any circle's circumference to its diameter, about three point one four one six. A sphere has surface area four pi r squared and volume four thirds pi r cubed [Equation 1.3-5]. A cylinder's volume is the area of its circular end times its length.

Notice how these formulas carry the square-cube law inside them. Surface area goes as the radius squared; volume goes as the radius cubed. So the ratio of surface to volume falls as things get bigger, which is why a large cup of coffee stays hot longer than a small one, and why cells must stay small to

exchange enough material through their surface.

Key ideas to carry forward

- Pythagoras: hypotenuse squared equals the sum of the other two squares
- Scale lengths by k: areas by k^2, volumes by k^3
- Circle: $C = 2\pi r$, $A = \pi r^2$. Sphere: $A = 4\pi r^2$, $V = (4/3)\pi r^3$
- Surface-to-volume ratio falls as size grows

Module 1.4

Trigonometry

Trigonometry connects angles to lengths. It began with surveyors and astronomers, and in physics it does two jobs: resolving forces and velocities into components, and describing anything that repeats, from pendulums to sound waves.

By the end of this module you can

- Use sine, cosine and tangent in right-angled triangles
- Measure angles in radians as well as degrees
- Relate the unit circle to the sine wave and describe amplitude, period and phase

SOH CAH TOA

In a right-angled triangle, pick one of the other angles and call it theta. The side across from it is the opposite, the side next to it is the adjacent, and the longest side is the hypotenuse. Sine of theta is opposite over hypotenuse; cosine is adjacent over hypotenuse; tangent is opposite over adjacent [Equation 1.4-1]. The old mnemonic SOH CAH TOA still works.

To find a length, rearrange: the opposite side equals the hypotenuse times sine theta. To find an angle, use an inverse function: theta equals the inverse sine of opposite over hypotenuse. On a calculator these are often labelled sine to the minus one, which means inverse, not one over sine.

Components

The most common use in physics is resolving. A force of ten newtons pulling at thirty degrees above the horizontal does two things at once: it pulls sideways with ten times cosine thirty, about eight point seven newtons, and lifts with ten times sine thirty, five newtons [Equation 1.4-2]. Splitting a quantity into perpendicular components turns one awkward diagonal problem into two easy straight-line ones, which is the key move in Course 3.

Radians

Degrees are a human convention: three hundred and sixty in a circle, inherited from Babylonian astronomy. Mathematics prefers radians. One radian is the angle at which the arc length equals the radius. A full circle is two pi radians, so one hundred and eighty degrees is pi radians [Equation 1.4-3]. Radians make many formulas simpler: the arc length is just radius times angle [Equation 1.4-4], and in calculus the derivative of sine is cosine only when angles are in radians. Check which mode your calculator is in before every session.

The unit circle and waves

Draw a circle of radius one. A point travelling around it at angle theta has horizontal position cosine theta and vertical position sine theta. This defines sine and cosine for any angle, not just those inside a triangle, and gives the identity sine squared plus cosine squared equals one [Equation 1.4-5], which is

Pythagoras in disguise.

Plot the height of that point against angle and you get the sine wave. As the point goes round and round, the wave repeats. A general wave is written as amplitude times sine of the quantity omega t plus phi [Equation 1.4-6]. The amplitude is the height of the peaks, the period is the time for one cycle, and the phase shifts the wave along. Tides, alternating current and sound all have this form.

Bridge from the social sciences

Seasonal adjustment in time-series data uses sine and cosine terms to model regular cycles, the same mathematics as tides and alternating current.

Key ideas to carry forward

- SOH CAH TOA in right triangles; inverse functions give angles
- Resolve vectors with cos for the adjacent component and sin for the opposite
- π radians = 180°; arc length $s = r\theta$ (θ in radians)
- Sine waves: amplitude, period and phase

Module 1.5

Probability and statistics

You already know more statistics than most science students. This module translates it into the language scientists use for measurement, uncertainty and chance, and fills in the probability rules underneath.

By the end of this module you can

- Apply the addition and multiplication rules, and conditional probability
- Use mean, standard deviation and the normal distribution to describe data
- Distinguish standard deviation from standard error, and read error bars

Probability rules

A probability is a number from zero, impossible, to one, certain. For equally likely outcomes it is the number of favourable outcomes divided by the total. Two rules do most of the work. For mutually exclusive events, the probability of one or the other is the sum [Equation 1.5-1]. For independent events, the probability of both is the product [Equation 1.5-2]. The chance of rolling two sixes is one sixth times one sixth, one in thirty-six.

Conditional probability asks: given that B has happened, how likely is A? It is the probability of both, divided by the probability of B [Equation 1.5-3]. Bayes' theorem turns this around and is the reason that a ninety-nine per cent accurate test for a rare disease still produces mostly false positives. You may already know this from evidence-based policy; in genetics, Course 6 uses exactly the same reasoning.

Describing data

The mean is the sum divided by the count. The standard deviation measures typical spread around the mean [Equation 1.5-4]. Many measured quantities follow the bell-shaped normal distribution, where about sixty-eight per cent of values lie within one standard deviation of the mean and about ninety-five per cent within two.

In science the spread usually comes from measurement noise rather than real differences between people. Measure the same pendulum's period twenty times and the readings scatter. Their mean is your best estimate; their standard deviation describes how much a single reading wobbles.

Standard error and error bars

The precision of the mean is better than that of a single reading. The standard error of the mean is the standard deviation divided by the square root of the number of readings [Equation 1.5-5]. Quadruple your readings and you halve the uncertainty in the mean. This is the law of large numbers at work, illustrated in the Visual Companion.

Error bars on a graph may show standard deviation, standard error or a confidence interval, and good papers say which. A rough rule: if two means differ by more than about two standard errors, the difference is unlikely to be chance. Physicists sometimes demand five standard deviations before announcing a discovery, which is why the Higgs boson took so long to confirm.

Bridge from the social sciences

Statistics in physics is mostly about measurement error rather than population variation, and physicists talk about 'sigma' where you might talk about p-values. A 2-sigma result corresponds roughly to $p \approx 0.05$; the 5-sigma particle-physics standard corresponds to about one in 3.5 million.

Key ideas to carry forward

- Exclusive events: add probabilities. Independent events: multiply
- Conditional probability: $P(A|B) = P(A \text{ and } B) \,/\, P(B)$
- Normal distribution: 68% within 1σ, 95% within 2σ
- Standard error = $\sigma/\sqrt{n}$; more data shrinks it slowly

Module 1.6

Vectors

Some quantities need only a size: mass, temperature, energy. Others also need a direction: velocity, force, displacement. These are vectors, and handling them properly is the gateway to physics.

By the end of this module you can

- Distinguish scalars from vectors and represent vectors in components
- Add vectors graphically and by components, and find magnitude and direction
- Calculate and interpret the dot product

Scalars and vectors

A scalar is a number with a unit: five kilograms, twenty degrees Celsius. A vector has a size, called its magnitude, and a direction: thirty metres per second heading north-east. On paper we draw vectors as arrows, the length showing magnitude and the arrowhead showing direction.

Distance is a scalar; displacement is a vector. Walk four kilometres east then three kilometres north and you have covered seven kilometres of distance, but your displacement is five kilometres in a north-easterly direction. The difference matters every time something changes direction.

Components

Any vector in a flat plane can be written as how far it goes in the x direction and how far in the y direction. These are its components, written in brackets: a equals four, one. From components, the magnitude follows by Pythagoras [Equation 1.6-1] and the direction by inverse tangent [Equation 1.6-2]. Going the other way, a vector of magnitude F at angle theta has components F cos theta and F sin theta, as you saw in Module 1.4.

Adding vectors

Graphically, vectors add tip to tail: draw the first, start the second at its tip, and the sum runs from the start of the first to the tip of the second. Algebraically it is even easier: add the components separately [Equation 1.6-3]. Four, one plus minus one, three equals three, four, whose magnitude is five.

Subtracting a vector means adding its reverse. Multiplying a vector by a number stretches it, and a negative number also flips it. In three dimensions everything carries over with a third component, z.

The dot product

There are two ways to multiply vectors. The one you need now is the dot product, which gives a scalar. In components, multiply matching components and add [Equation 1.6-4]. Geometrically, it equals the product of the magnitudes times the cosine of the angle between them [Equation 1.6-5]. So the dot product is zero when vectors are perpendicular, and largest when they point the same way.

Physics uses this directly. Work done by a force is the dot product of force and displacement. Push a trolley straight ahead and all your effort counts. Push at an angle and only the component along the motion does work. The other kind of product, the cross product, gives a vector and appears with magnetism in Course 3.

Key ideas to carry forward

- Vectors have magnitude and direction; scalars only magnitude
- Magnitude = $\sqrt{(x^2 + y^2)}$; direction from $\tan^{-1}(y/x)$
- Add vectors by adding components
- $a \cdot b = a_x b_x + a_y b_y = |a||b| \cos \theta$; zero when perpendicular

Module 1.7

Mathematical thinking

Mathematics is not only calculation. It is a way of reasoning: stating assumptions, deducing consequences, and building models that are simple enough to solve but faithful enough to be useful. This module is about that way of thinking.

By the end of this module you can

- Distinguish definitions, assumptions, theorems and proofs
- Follow simple direct proofs and proofs by contradiction
- Describe the modelling cycle and judge when a model's assumptions break down

Statements, definitions and proof

A mathematical argument starts from definitions and agreed assumptions, called axioms, and reaches conclusions by steps that cannot be disputed. A proven statement is a theorem. Unlike a scientific theory, which is always open to revision by new evidence, a theorem stays true forever once proved, given its assumptions.

A direct proof chains implications. To show that the sum of two even numbers is even: write them as two m and two n; their sum is two times m plus n, which is even by definition [Equation 1.7-1]. A single example proves nothing about all cases, but a single counter-example disproves a general claim.

Proof by contradiction

Sometimes it is easier to assume the opposite and show it leads to nonsense. The classic example, known to the ancient Greeks, is that the square root of two cannot be written as a fraction. Suppose it could, as a over b in lowest terms. Then a squared equals two b squared [Equation 1.7-2], so a squared is even, so a is even. Write a as two k; then b squared equals two k squared, so b is even too. But we said the fraction was in lowest terms. Contradiction. So root two is irrational.

Models and modelling

Science uses mathematics to build models: simplified descriptions that capture what matters. The modelling cycle goes from a real question, through simplifying assumptions, to equations, to a solution, and back to reality to check whether the answer makes sense. If it doesn't, you revise the assumptions.

Every model leaves things out. Projectile motion ignores air resistance; the ideal gas law ignores the size of molecules; exponential population growth ignores limits on food. The skill is knowing when those omissions matter. A good scientist can tell you not only what a model predicts but where it will fail.

Two habits help. First, check limiting cases: what does your formula predict if a quantity goes to zero or becomes very large? Second, check units and orders of magnitude. If a model predicts a raindrop falling faster than a jet, the assumptions need another look.

Bridge from the social sciences

You will recognise the modelling cycle from quantitative social science, including the debate about whether a model is meant to predict or to explain. The difference in physics is that models are often derived from first principles and tested against very precise data.

Key ideas to carry forward

- Theorems are proved from definitions and axioms; they do not get overturned
- One counter-example disproves a general claim; one example proves nothing
- Proof by contradiction: assume the opposite, derive nonsense
- Every model has assumptions; know where they break